

# JEE MAIN REVISION MASTER: TRIGONOMETRIC RATIO AND IDENTITES

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 69 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                                                   | Frequency | Category              |
|---------------|--------------------------------------------------------------------------------|-----------|-----------------------|
| Very High     | Trigonometric Ratio And Identites — Standard Calculations & Problem Archetypes | 54 PYQs   | Quantitative Analysis |
| High          | Trigonometric Ratio And Identites — Applied Cases, Reactions & Traps           | 10 PYQs   | Applied Topics        |
| Moderate      | Trigonometric Ratio And Identites — Core Principles & Governing Laws           | 6 PYQs    | Core Principle        |

## 2. Master Formula Reference Card

| Concept / Topic                                                                | Mathematical Formulation & Governing Equations                                                                                          |
|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Trigonometric Ratio And Identites — Standard Calculations & Problem Archetypes | $\operatorname{cosec}10^{\circ}-\sqrt{3}\sec10^{\circ}, \quad 3\sin(\alpha+\beta)=2\sin(\alpha-\beta),$<br>$\quad \frac{1}{2} \times 4$ |
| Trigonometric Ratio And Identites — Applied Cases, Reactions & Traps           | $=\frac{\sin^2A+\sin A\cos A+\cos^2A}{\sin A\cos A}, \quad =\frac{1}{16}, \quad$<br>$3\sin(\alpha+\beta)=2\sin(\alpha-\beta)$           |
| Trigonometric Ratio And Identites — Core Principles & Governing Laws           | $\sin x+\sin^2x=1, \quad$ is equal to $\rule{1cm}{0.4pt}, \quad$<br>$=\frac{1}{16}[\sin10^{\circ}(\cos20^{\circ})-\frac{1}{2}]$         |

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Trigonometric Ratio And Identities — Standard Calculations & Problem Archetypes (Very High Yield — 54 PYQs)

**Core Principle:** Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Trigonometric Ratio And Identities.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

**Representative Source PYQs:**

- [2026 Morning Shift] Q#1 (MCQ): "Let  $\tan A, \tan B$ , where  $A, B \in (-\pi/2, \pi/2)$ , be the roots of the quadratic equation  $x^2 - 2x - 5 = 0$ . Then  $20\sin^2\{ \dots \}$ " [Ans: C]
- [2026 Morning Shift] Q#1 (MCQ): "The value of  $\frac{\sqrt{3}\operatorname{cosec}20^\circ - \sec20^\circ \cos20^\circ \cos40^\circ \cos60^\circ \cos80^\circ}{\dots}$  is equal to..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (Numerical): "If  $A = \frac{\sin3^\circ \cos9^\circ + \sin9^\circ \cos27^\circ + \sin27^\circ \cos81^\circ}{\dots}$  and  $B = \tan81^\circ - \tan3^\circ$ " [Ans: 2]

### 3.2 Trigonometric Ratio And Identities — Applied Cases, Reactions & Traps (High Yield — 10 PYQs)

**Core Principle:** Special cases, experimental observations, and conceptual traps designed to test deep understanding of Trigonometric Ratio And Identities.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

**Representative Source PYQs:**

- [2026 Evening Shift] Q#2 (MCQ): "Let  $P = \{ \theta \in [0, 4\pi] : \tan^2 \theta \neq 1 \text{ and } S = a \in \mathbb{R} : 2(\cos^8 \theta - \sin^8 \theta) \sec 2\theta = a^2, \theta \in P \}$ . Then  $n(S)$ ..." [Ans: N/A]
- [2024 Evening Shift] Q#4 (MCQ): "For  $\alpha, \beta \in (0, \pi/2)$ , let  $3\sin(\alpha + \beta) = 2\sin(\alpha - \beta)$  and a real number  $k$  be such that  $\tan \alpha = k \tan \beta$ . Then..." [Ans: N/A]
- [2022 Evening Shift] Q#2 (MCQ): " $2\sin((\pi/22))\sin((3\pi/22))\sin((5\pi/22))\sin((7\pi/22))\sin((9\pi/22))$  is equal to..." [Ans: N/A]

### 3.3 Trigonometric Ratio And Identities — Core Principles & Governing Laws (Moderate Yield — 6 PYQs)

**Core Principle:** Fundamental definitions, core postulates, and governing physical/chemical relations in Trigonometric Ratio And Identities.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Trigonometric Ratio And Identities.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Trigonometric Ratio And Identities PYQs.

**Representative Source PYQs:**

- **[2026 Evening Shift] Q#3 (Numerical):** "Let  $\cos(\alpha+\beta)=-1/10$  and  $\sin(\alpha-\beta)=3/8$ , where  $0<\alpha<(\pi/3)$  and  $0<\beta<(\pi/4)$ . If ..." [Ans: 20]
- **[2025 Evening Shift] Q#3 (MCQ):** "If  $\sin x + \sin^2 x = 1$ ,  $x \in (0, (\pi/2))$ , then  $(\cos^2 12x + \tan^2 12x) + 3(\cos^2 10x + \tan^2 10x + \cos^2 8x + \tan^2 8x) + (\cos^2 6x + \tan^2 6x) \dots$ " [Ans: N/A]
- **[2024 Evening Shift] Q#3 (MCQ):** "The number of solutions, of the equation  $e^{\sin x} - 2e^{-\sin x} = 2$ , is ..." [Ans: N/A]

## 4. Benchmark Illustrative PYQs with Master Solutions

### Benchmark Problem 1: JEE Main 2026 (Online) 5th April Morning Shift (Q#1)

Let  $\tan A, \tan B$ , where  $A, B \in (-\pi/2, \pi/2)$ , be the roots of the quadratic equation  $x^2 - 2x - 5 = 0$ . Then  $20\sin^2((A+B)/2)$  is equal to:

- (A)  $10 + \sqrt{10}$
- (B)  $10 - 2\sqrt{10}$
- (C)  $10 - 3\sqrt{10}$  [Correct]
- (D)  $10 - \sqrt{10}$

**Correct Answer: C**

**Solution Analysis:** Let the roots of the equation  $x^2 - 2x - 5 = 0$  be  $\tan A$  and  $\tan B$ . So, by using relations between roots and coefficients,  $\tan A + \tan B = 2$  and  $\tan A \tan B = -5$ . We have to find  $20\sin^2((A+B)/2)$ . First, find  $\tan(A+B)$  using the formula  $\tan(A+B) = (\tan A + \tan B) / (1 - \tan A \tan B)$ . Substituting the values:  $\tan(A+B) = (2 / (1 - (-5))) = (2/6) = (1/3)$ . Now let  $\theta = A+B$ . Then  $\tan \theta = (1/3)$ . We need  $\sin^2(\theta/2)$ . Using the identity  $\sin^2(\theta/2) = (1 - \cos \theta) / 2$ , we first find  $\cos \theta$ . Since  $\tan \theta = (1/3)$ , take a right triangle with

### Benchmark Problem 2: JEE Main 2026 (Online) 22nd January Evening Shift (Q#3)

Let  $\cos(\alpha+\beta)=-1/10$  and  $\sin(\alpha-\beta)=3/8$ , where  $0<\alpha<(\pi/3)$  and  $0<\beta<(\pi/4)$ . If  $\tan 2\alpha = \frac{3(1-\sqrt{5})}{\sqrt{11}(s+\sqrt{5})}$ ,  $r, s \in \mathbb{N}$ , then  $r+s$  is equal to \_\_\_\_.

**Correct Answer: 20**

**Solution Analysis:**  $\tan 2\alpha = \tan[(\alpha+\beta) + (\alpha-\beta)]$   
 $\tan 2\alpha = \frac{\tan(\alpha+\beta) + \tan(\alpha-\beta)}{1 - \tan(\alpha+\beta) \cdot \tan(\alpha-\beta)}$   
 $\tan 2\alpha = \frac{-\sqrt{99} + \frac{3\sqrt{55}}{1 - (\sqrt{99})(\frac{3\sqrt{55}}{10})}}{1 - \frac{3\sqrt{11} + \frac{3\sqrt{5}}{\sqrt{11}}}{1 + \frac{9\sqrt{11}\sqrt{5}}{10}}} \times \sqrt{11}$   
 $\tan 2\alpha = \frac{3(1-11\sqrt{5})}{\sqrt{11}(9+\sqrt{5})}$   $r=11, s=9$   $r+s=20$